

WELL COMPLETION REPORT (SYNTHETIC)

OIL-RDS-003 - Rudrasagar-003

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000030
Field / District	Rudrasagar / Sivasagar
Block	PEL-ASSAM-06
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2013-06-27 / 2013-10-26
Measured / TVD	3256 m / 3215 m
Trajectory	Directional; inc 41.2 deg; az 296.0 deg
Formation Interval	1928-2772 m
Lithology	Calcareous shale and siltstone
Pore Pressure Reference	3846 psi
Fracture Gradient	14.91 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	High (61.6/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	814	Surface
9 5/8 in casing	1888	Intermediate
7 in liner/casing	2865	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2013-07-24 895 m	Kick	Critical	Flow-out exceeded flow-in and gas units increased. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Maintain mud-weight window using offset pressure evidence.
2013-08-11 1359 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Maintain mud-weight window using offset pressure evidence.
2013-08-16 1497 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2013-09-05 2004 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2013-10-18 3045 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2013-10-20 3104 m	Cementing Issue	High	Cement returns were below planned percentage. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Plan conservative displacement rate in weak formations.
2013-10-21 3127 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.

Completion Summary

The synthetic well reached 3256 m MD in the Kopili interval. The NWIS knowledge record links 16 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.